



Broadband High Power Amplifier Module (SSPA)

REV. 2.0_2023.02.28

6-18GHz/30Watt/Module

Model Number: KB60180M45B

The model KB60180M45B is a multi-octave high power amplifier operating between 6 GHz and 18 GHz and offering a wide dynamic Range with 30 Watts typical saturated power. The employment of gallium nitride (GaN) and chip-and-wire technology in manufacturing ensures this module state-of-the-art power performance with excellent power-to-volume ratio. It is ideal for jamming, EMC, test and measurement applications.

FEATURES:

- Small Size and light weight
- Instantaneous ultra-broadband
- 50 Ohms input and Output matched
- Built-in control and protection circuits

ELECTRICAL SPECIFICATIONS @ +28.0VDC, 25°C, 50Ω

Parameter	Symbol	Minimum	Typical	Maximum	Units
Operating Frequency	BW	6000		18000	MHz
RF Output Power @PSAT	PSAT		30		Watt
Power Gain @PSAT	Gp		45		dB
Power Gain Flatness	Δ Gp		± 2		dB
Input Return Loss	S ₁₁		-10		dB
Harmonics @20W	H		-13		dBc
Spurious Signals	Spur		-60		dBc
In/Output Impedance			50		Ω
Operating Voltage	V _{DC}	26	28	30	Volt
DC Current @30W	I _{DD}		9		Amp
Switching Time @1kHz TTL	T _{on/off}		1		uSec

MECHANICAL SPECIFICATIONS

Parameter	Value	Units	Limits
Dimensions	150x90x30 [5.90x3.54x1.18]	mm [inch]	Maximum
Weight	1.0 [2.2]	kg [lbs]	Maximum
RF Connectors Input	SMA, Female		
RF Connectors Output	SMA, Female		
DC Interface Connector	J29A-15ZK		
Cooling	External Heat sink Required (Not Supplied)		

ENVIRONMENTAL CHARACTERISTICS (Design to Meet)

Parameter	Minimum	Typical	Maximum	Units	Notes
Operating Temperature	-20		55	°C	
Non-operating Temperature	-25		65	°C	Storage
Relative Humidity (non-condensing)			95	%	

Absolute Maximum Rating

Input RF drive level without damage	+5 dBm	Maximum
Load VSWR @ POUT =20W	3:1 @ all load phase & amplitude continuous More than 4:1 may cause PA damage	
Over Temperature	85°C @ heat sink [restored @ 60°C]	

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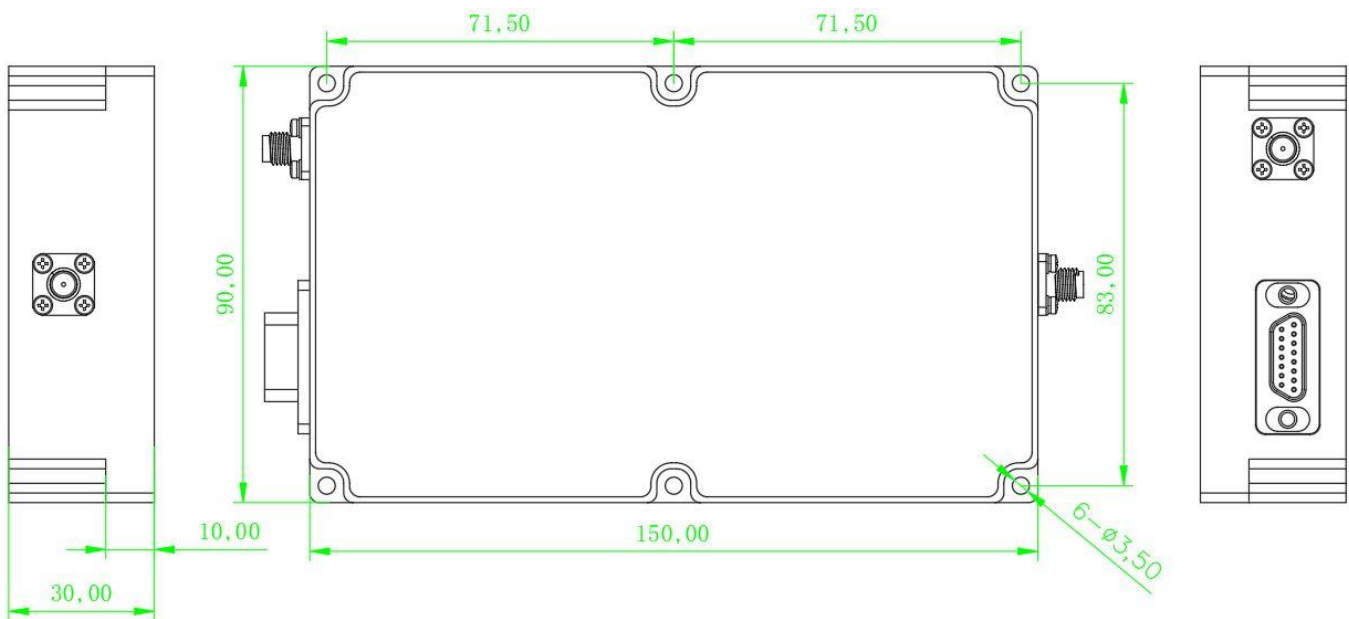
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DC INTERFACE CONNECTOR

Pin #	Description	Specifications
1-4	VDD	+28V
5-9	GND	Ground
10	ENABLE	Amplifier Enable: TTL Logic High (3.3V)
11	TEMP SENSE	TTL Logic High (3.3V) when temperature was exceed limit value
12	CURRENT SENSE	Current Monitoring
13-15	NC	No electrical connection

OUTLINE DRAWING [All dimensions in mm]





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TYPICAL PERFORMANCE PLOTS (for reference only)

